

Chapter 11:

Reactions Involving Gases

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11.1. Introduction

- ✓ In Chapter 8, it was seen that the absence of interatomic forces between the atoms of ideal gases causes the enthalpy of mixing of ideal gases to be zero. This situation represents one extreme of a range of possible situations.
- ✓ Toward the other extreme of this range is the situation in which gases are mixed which exhibit marked chemical affinity for one another.
- ✓ For example, considerable heat is released when gaseous hydrogen and oxygen are mixed in the presence of a catalyst.

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- ✓ The thermodynamics of such a system can be treated in either of two ways:
 1. The mixture can be considered to be a highly nonideal mixture of H_2 and O_2 , the thermodynamic equilibrium state of which, at a given temperature and pressure, can be defined in terms of the fugacities of the components H_2 and O_2 .
 2. It can be considered that the H_2 and O_2 have reacted with one another to some extent to produce the product species H_2O .

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- ✓ In the latter case, if the pressure of the system is low enough, the equilibrium state at the given temperature can be defined in terms of the partial pressures exerted by the three species, H_2 , O_2 , and H_2O , occurring in the system.
- ✓ Although both treatments are thermodynamically equivalent, the latter is the more convenient and practical.
- ✓ As with any constant-pressure, constant-temperature system, the equilibrium state is that in which the Gibbs free energy of the system has its minimum possible value.

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- ✓ If the gases which are present initially in the system react to form distinct product species, the total change in the Gibbs free energy of the system comprises a contribution arising from
 - The *change* in Gibbs free energy due to the chemical reaction
 - The *mixing* of the appearing product gases with the remaining reactant gases
- ✓ Knowledge of the variation of this total change in the Gibbs free energy with composition (which ranges from the pure unmixed reactant gases to the pure unmixed product gases) allows for the determination of the equilibrium state in any system of reactive gases.

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- ✓ This determination is facilitated by the introduction of the **equilibrium constant for the reaction**, and it will be seen that the relation between this constant and the standard Gibbs free energy change for the reaction is one of the more important relationships in reaction equilibrium thermodynamics.

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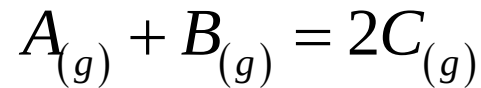
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- ✓ Consider the reaction



- ✓ occurring at constant temperature T and constant pressure P .
- ✓ At any instant during the reaction, the total Gibbs free energy of the system is

$$G' = n_A \bar{G}_A + n_B \bar{G}_B + n_C \bar{G}_C$$

where n_A , n_B , and n_C are, respectively, the numbers of moles of A, B, and C present in the reaction system at that instant, and $\bar{G}_A$, $\bar{G}_B$ and $\bar{G}_C$ are, respectively, the partial molar Gibbs free energies of A, B, and C in the gas mixture which occurs at that instant.

11.2. Reaction Equilibrium in a Gas Mixture and the Equilibrium Constant

- ✓ It is of interest to determine the values of n_A , n_B , and n_C which minimize the value of G' in Equation 11.1, since this state of minimum Gibbs free energy is the equilibrium state of the system at the given temperature and pressure.
- ✓ That is, once the chemical reaction between A and B has proceeded to the extent that the Gibbs free energy of the system has been minimized, the reaction at the macroscopic level appears to stop.
- ✓ On the microscopic level, however, reaction equilibrium occurs when the rate of the chemical reaction proceeding from right to left equals that of the reaction proceeding from left to right.

11.2. Reaction Equilibrium in a Gas Mixture and the Equilibrium Constant

- ✓ This is a *dynamic equilibrium*: that is, reactions still occur when the system is at equilibrium.
- ✓ The stoichiometry of the reaction allows the numbers of moles of all the species present at any instant to be expressed in terms of the number of moles of any one of the species.
- ✓ Starting with 1 mole of A and 1 mole of B (i.e., 2 moles of gas), as 1 atom of A reacts with 1 atom of B to produce 2 molecules of C, then, at any time during the reaction,

$$n_A = n_B$$

- ✓ and

$$n_C = 2 - n_A - n_B = 2(1 - n_A)$$

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✓ Equation 11.1 can thus be written as

$$G' = n_A \bar{G}_A + n_B \bar{G}_B + 2(1 - n_A) \bar{G}_C$$

✓ From Equation 8.15:

$$\bar{G}_i = G_i^\circ + RT \ln P + RT \ln X_i$$

✓ and

$$X_A = \frac{n_A}{2} \quad X_B = \frac{n_A}{2} \quad X_C = \frac{2(1 - n_A)}{2} = (1 - n_A)$$

✓ substitution of which gives

$$G' = n_A \left(\underline{G_A^\circ + G_C^\circ - 2G_C^\circ} \right) + \underline{2G_C^\circ} + 2RT \ln P \\ + 2RT \left[n_A \ln \left(\frac{n_A}{2} \right) + (1 - n_A) \ln (1 - n_A) \right]$$

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✓ or
$$\underline{G'} - 2G_C^\circ = n_A \left(\underline{-\Delta G^\circ} \right) + 2RT \ln P$$
$$+ 2RT \left[n_A \ln \left(\frac{n_A}{2} \right) + (1 - n_A) \ln (1 - n_A) \right]$$

- ✓ The term $\Delta G^\circ = 2\Delta G_C^\circ - \Delta G_A^\circ - \Delta G_B^\circ$ is the *standard Gibbs free energy change* for the chemical reaction at the temperature T .
- ✓ The standard Gibbs free energy change (ΔG°) for any reaction is the difference between the sum of the Gibbs free energies of the reaction products in their standard states and the sum of the Gibbs free energies of the reactants in their standard states.

$$\Delta G^\circ = \Delta G_{\text{products}}^\circ - \Delta G_{\text{reactants}}^\circ$$

11.2. Reaction Equilibrium in a Gas Mixture and the Equilibrium Constant

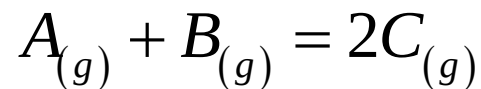
- ✓ In the present case, ΔG° is the difference between the Gibbs free energy of 2 moles of C at 1 atm pressure and the temperature T , and 1 mole of A and 1 mole of B, each at 1 atm pressure and the temperature T .
- ✓ If the total pressure of the system is 1 atm, then Equation 11.2 simplifies to

$$G' - 2G_C^\circ = n_A(-\Delta G^\circ) + 2RT \left[n_A \ln \left(\frac{n_A}{2} \right) + (1 - n_A) \ln (1 - n_A) \right]$$

- ✓ The left-hand side of Equation 11.3 is the difference between the Gibbs free energy of the 2-mole system when $n_A = n_B$ and the Gibbs free energy of the system when it consists of 2 moles of C.

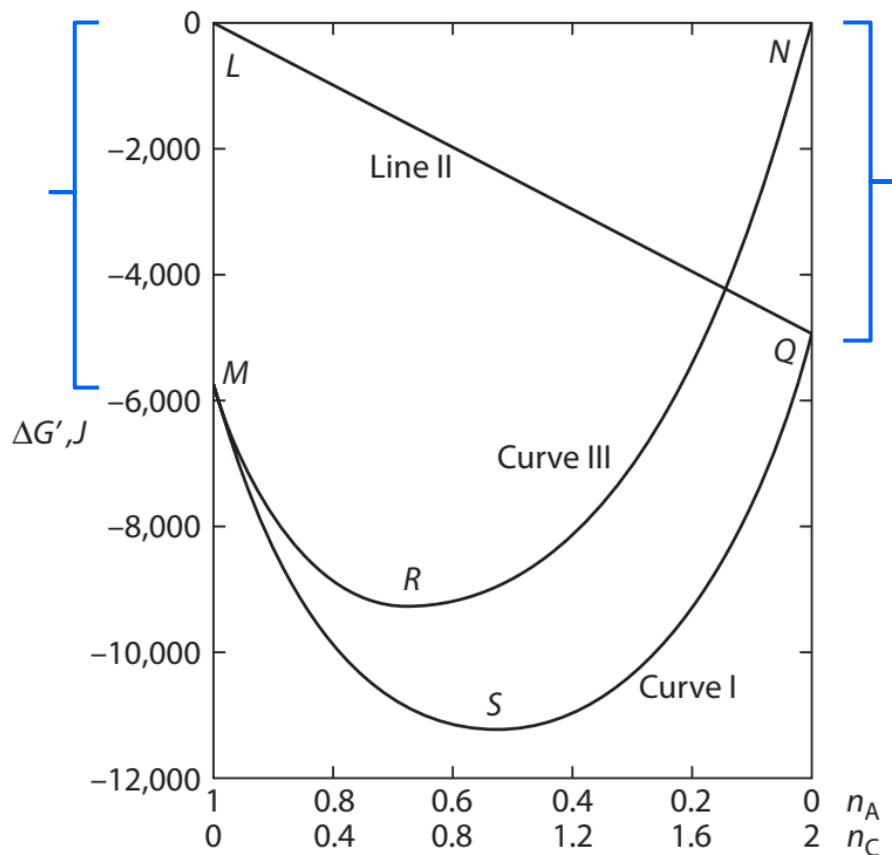
11.2. Reaction Equilibrium in a Gas Mixture and the Equilibrium Constant

- ✓ This difference is determined by two factors:
 1. The *change* in the Gibbs free energy due to the *chemical reaction*—that is, due to the disappearance of the reactants and the appearance of the products.
 2. The *decrease* in the Gibbs free energy caused by *mixing* of the gases.
- ✓ Figure 11.1 is drawn for the reaction.



at 500 K and 1 atm pressure.

11.2. Reaction Equilibrium in a Gas Mixture and the Equilibrium Constant



- ✓ point L ($n_A = 1$, $n_B = 1$ before mixing) is located at $\Delta G' = 0$, and the point Q ($n_C = 2$) is located at $\Delta G' = -5000$ J.
- ✓ The point M represents the decrease in the Gibbs free energy due to mixing before any chemical reaction occurs;

Figure 11.1 The variations of the contributions to the decrease in the Gibbs free energy due to chemical reaction (line II), the contribution to the decrease in the Gibbs free energy due to gas mixing (curve III), with the extent of the reaction $A_{(g)} + B_{(g)} = 2C_{(g)}$, for which $\Delta G^\circ = -5000$ J at 500 K. Curve I is the sum of curves II and III.

11.2. Reaction Equilibrium in a Gas Mixture and the Equilibrium Constant

- ✓ ΔG_{500}° for the reaction is taken as -5000 J . If the reference for the Gibbs free energy is arbitrarily chosen as $(\Delta G_A^\circ + \Delta G_B^\circ) = 0$, then $2\Delta G_C^\circ = -5000 \text{ J}$.
- ✓ In Figure 11.1, the ordinate $\Delta G'$ is plotted as the difference between the Gibbs free energy of the system containing n_A moles of A and the Gibbs free energy of the system comprising 1 mole of A and 1 mole of B before the mixing of A and B occurs.

11.2. Reaction Equilibrium in a Gas Mixture and the Equilibrium Constant

- ✓ The point M represents the decrease in the Gibbs free energy due to mixing of 1 mole of A and 1 mole of B before any chemical reaction between the two occurs; that is, from Equation 8.20:

$$\Delta G'(L \rightarrow M) = \sum_i n_i RT \ln \left(\frac{p_i}{P_i} \right) = RT \left[n_A \ln \left(\frac{p_A}{P_A} \right) + n_B \ln \left(\frac{p_B}{P_B} \right) \right]$$

- ✓ But $n_A = n_B = 1$, and

$$P_A = P_B = P_{\text{mixture}} = 1$$

- ✓ in which case

$$p_A = p_B = \frac{1}{2}$$

11.2. Reaction Equilibrium in a Gas Mixture and the Equilibrium Constant

✓ Thus,

$$\Delta G(L \rightarrow M) = 8.3144 \times 500 \times 2 \times \ln 0.5 = -5763 J$$

✓ Substituting this into Equation 11.3 we obtain

$$G' + 5000 = (1 \times 5000) - 5763 J$$

✓ and hence

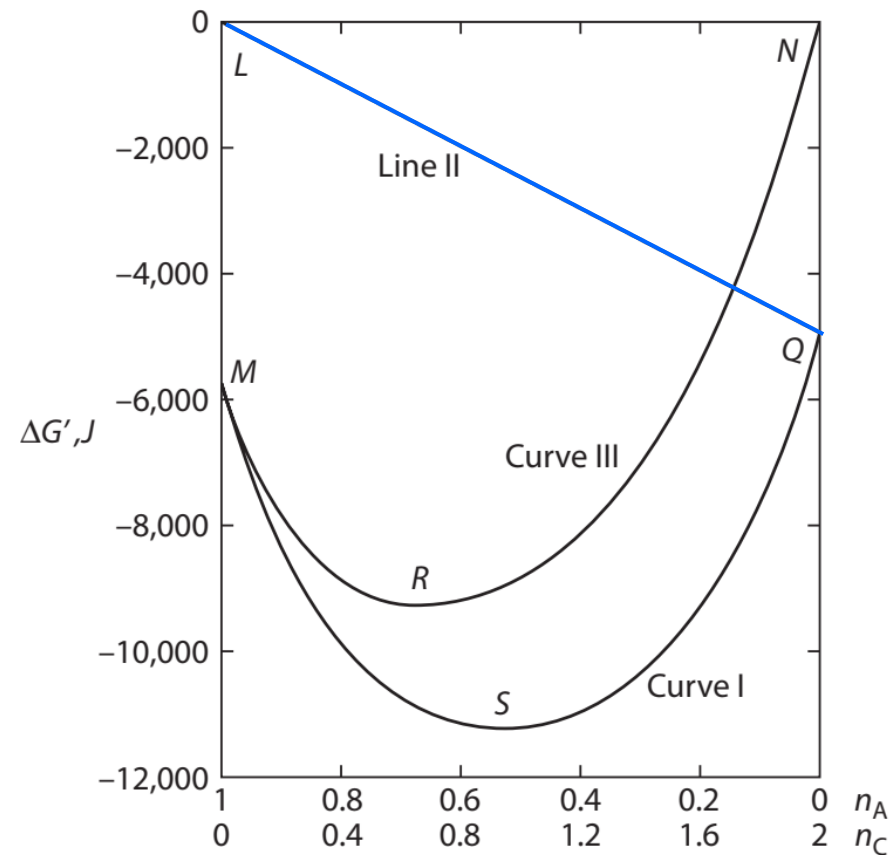
$$G' = -5763 J$$

✓ Thus, at M ,

$$\Delta G' = G' - (G_A^\circ + G_B^\circ) = -5763 J$$

11.2. Reaction Equilibrium in a Gas Mixture and the Equilibrium Constant

- ✓ **Curve I** represents the variation of $\Delta G'$ with n_A . It is obtained as the **sum** of line II and curve III. The magnitude of the **contribution to the change** in the **total Gibbs free energy** due to **gas mixing** is **greatest** at the **composition R**. (**Curve III**)
- ✓ The magnitude of the **chemical reaction** contribution to the decrease in the Gibbs free energy of the system **increases linearly** with increasing n_C , (**Line II**)



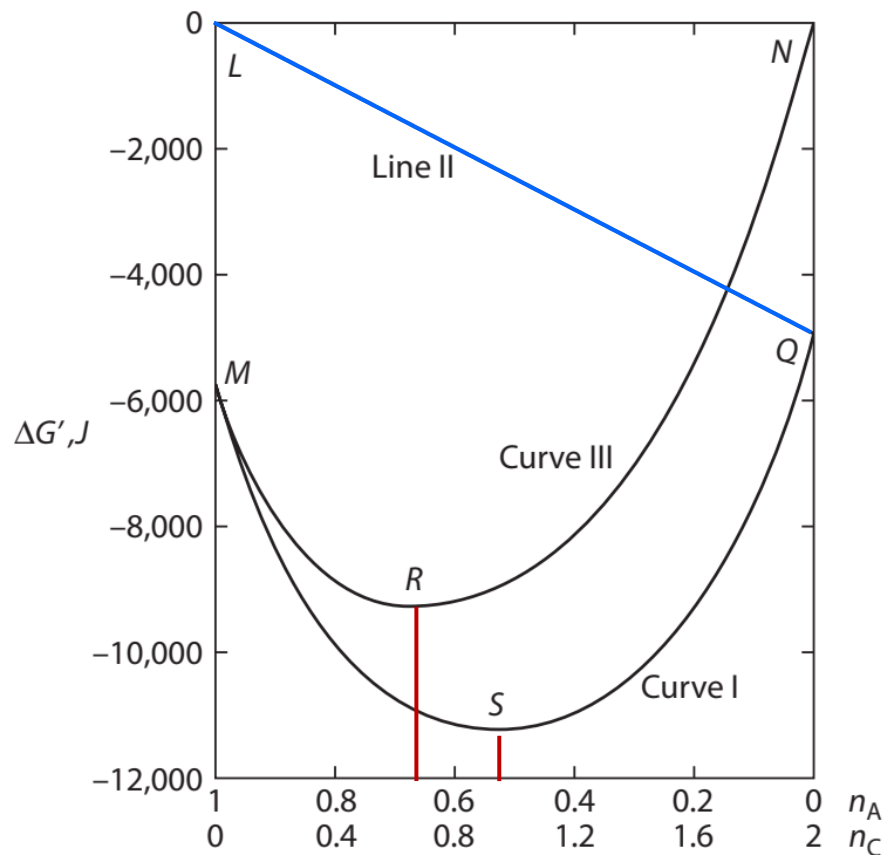
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- ✓ R occurs at that composition of the gas mixture which permits maximum randomization of the system.
- ✓ Further chemical reaction, which takes the composition of the system beyond R , decreases the magnitude of the gas-mixing contribution, since further increase in n_C , at the expense of n_A and n_B , decreases the randomness of the system.
- ✓ Eventually, the composition S is reached, in which state the sum of the two contributions to the decrease in the Gibbs free energy is a maximum.

11.2. Reaction Equilibrium in a Gas Mixture and the Equilibrium Constant

- ✓ If chemical reaction continued beyond S , then, since the decrease in **line II** is smaller than the increase in **curve III**, the total Gibbs free energy of the system would increase.
- ✓ The **composition S** is thus that at which the Gibbs free energy of the system has its minimum value and is hence the **equilibrium state**.

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composition $R \rightarrow S$, $\Delta G \downarrow$

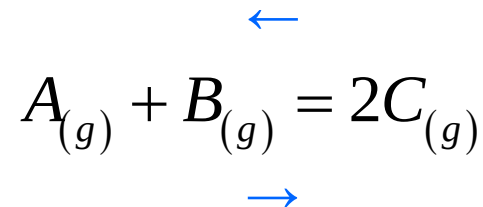


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- ✓ The position of the minimum in **curve I** is fixed by the criterion that, at the minimum,

$$\left(\frac{\partial G'}{\partial n_A} \right)_{T,P} = 0$$

- ✓ and since

$$G' = n_A \bar{G}_A + n_A \bar{G}_B + 2(1 - n_A) \bar{G}_C$$

- ✓ and including the use of the Gibbs–Duhem equation,*

$$\left(\frac{\partial G'}{\partial n_A} \right) = \bar{G}_A + \bar{G}_B - 2\bar{G}_C = 0$$

- ✓ That is, the **criterion** for **reaction equilibrium** is

$$\bar{G}_A + \bar{G}_B = 2\bar{G}_C$$

11.2. Reaction Equilibrium in a Gas Mixture and the Equilibrium Constant

- ✓ Equation 11.4 can be written as

$$G_A^\circ + RT \ln p_A + G_B^\circ + RT \ln p_B = 2G_C^\circ + 2RT \ln p_C$$

- ✓ where p_A , p_B , and p_C are, respectively, the partial pressures of A, B, and C which occur at reaction equilibrium.
- ✓ Rearrangement of Equation 11.5 gives

$$2G_C^\circ - G_A^\circ - G_B^\circ = -RT \ln \frac{p_C^2}{p_A p_B}$$

- ✓ or

$$\Delta G^\circ = -RT \ln \left(\frac{p_C^2}{p_A p_B} \right)$$

11.2. Reaction Equilibrium in a Gas Mixture and the Equilibrium Constant

$$\Delta G^\circ = -RT \ln\left(\frac{p_C^2}{p_A p_B}\right)$$

- ✓ The quotient of the equilibrium partial pressure of the reactants and products occurring as the logarithmic term in Equation 11.6 is termed the *equilibrium constant for the reaction*, K_p ; that is,

$$\left(\frac{p_C^2}{p_A p_B}\right)_{eq} = K_p$$

- ✓ and hence,

$$\Delta G^\circ = -RT \ln K_p$$

- ✓ Since ΔG° is a function only of temperature, it follows from Equation 11.8 that K_p is a function only of temperature.

11.2. Reaction Equilibrium in a Gas Mixture and the Equilibrium Constant

✓ For the example used in Figure 11.1,

$$\ln K_p = -\frac{\Delta G^\circ}{RT} = \frac{5000}{8.3144 \times 500} = 1.203$$

✓ Therefore,

$$K_p = 3.329$$

✓ Now,

$$K_p = \frac{p_C^2}{p_A p_B} = \frac{X_C^2 P^2}{X_A P X_B P} = \frac{X_C^2}{X_A X_B}$$

✓ and since $X_A = X_B = n_A/2$ and $X_C = (1 - n_A)$,

$$= \frac{(1 - n_A)^2}{\left(\frac{n_A^2}{4}\right)} = 3.329$$

11.2. Reaction Equilibrium in a Gas Mixture and the Equilibrium Constant

✓ Thus,

$$n_A = 0.523 \text{ (the other solution, } n_A = 11.4, \text{ is nonphysical)}$$

✓ The minimum in curve I in Figure 11.1 occurs at $n_A = n_B = 0.523$ and

$$n_C = 2 - 2 \times (0.523) = 0.954$$

✓ The chemical reaction $A + B = 2C$ has proceeded to 47.7% completion.

✓ If the temperature T was such that ΔG° for the reaction was zero, there would be no chemical reaction contribution to $\Delta G'$, and the variation of $\Delta G'$ with n_A would be given by curve III in Figure 11.1; that is, the criterion for reaction equilibrium would be the maximization of the configurational entropy of the system, which occurs at the composition R .

11.2. Reaction Equilibrium in a Gas Mixture and the Equilibrium Constant

$$\Delta G^\circ = -RT \ln K_p$$

- ✓ From Equation 11.8, if $\Delta G^\circ = 0$, then $K_p = 1$, and thus,

$$1 = \frac{4(1 - n_A)^2}{n_A^2} \quad \text{and} \quad n_A = \frac{2}{3}$$

- ✓ The minimum in curve III occurs at $n_A = n_B = n_C = \frac{2}{3}$, and thus, maximum entropy in the system occurs when all three species are present in equal amounts.
- ✓ It is important to note that the minimum in curve I in Figure 11.1, in representing the equilibrium state of the system at $P = 1$ atm and 500 K, is the only point on curve I which has any significance within the scope of classical equilibrium thermodynamics.

11.2. Reaction Equilibrium in a Gas Mixture and the Equilibrium Constant

- ✓ The point **S** is the only state on the **curve I** which lies on the equilibrium surface in P - T -composition space for the fixed values of $P = 1 \text{ atm}$ and $T = 500 \text{ K}$. Any change in P and/or T changes the curves II and III, which in turn produces a “**new**” **curve I**.
- ✓ This **new curve I** has a different minimum and hence represents a new equilibrium state. This is discussed further in the following section.

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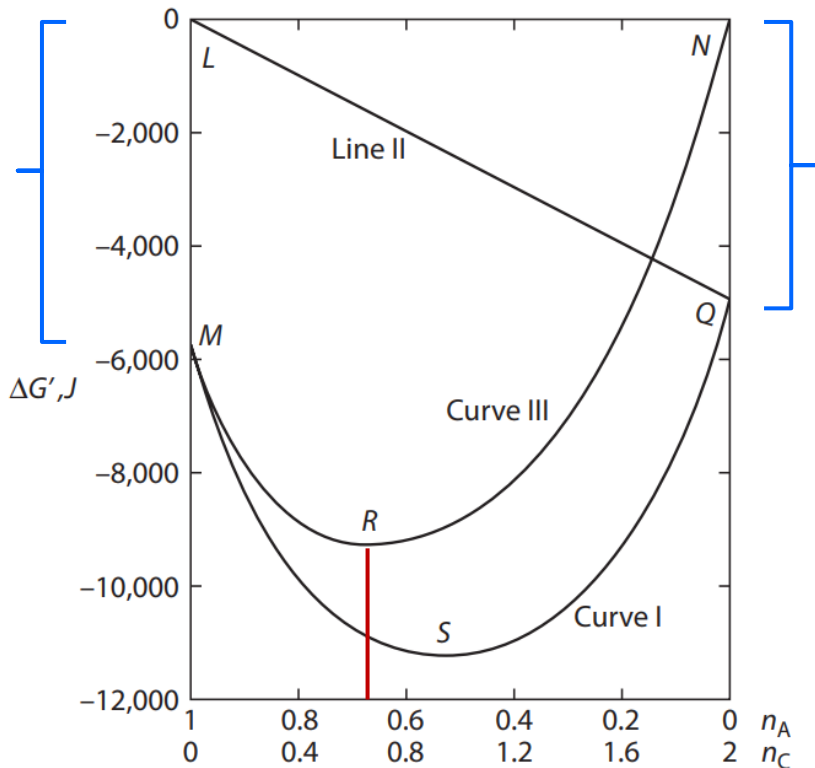
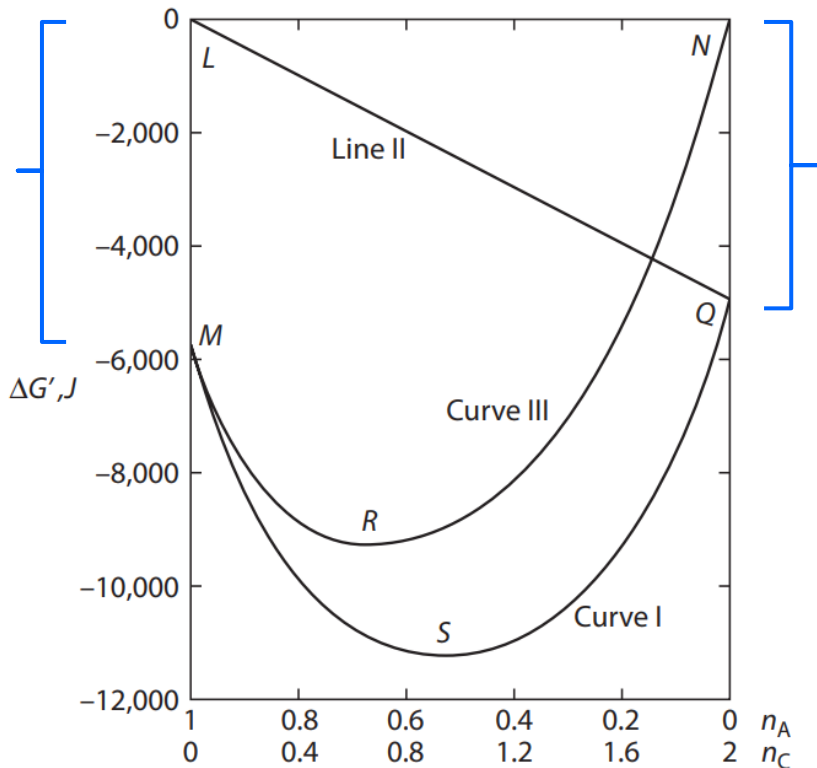


Figure 11.1

The decrease in ΔG due to **chemical reaction** (**line II**), due to gas **mixing** (**curve III**), Curve I is the sum of curves II and III.

- ✓ The position of the **minimum** in **curve I** in Figure 11.1 is determined by the difference between the lengths of **LM** and **NQ**.
- ✓ The length **LM** is the **decrease** in the Gibbs free energy caused by **mixing** of the gases **before** the reaction begins, and
- ✓ the length **NQ** is the standard Gibbs free energy change, ΔG° , for the **reaction**.

11.3. The Effect of Temperature on the Equilibrium Constant



- ✓ The **lengths** of both of these lines are dependent on temperature.

$$LM = 2RT \ln 0.5$$

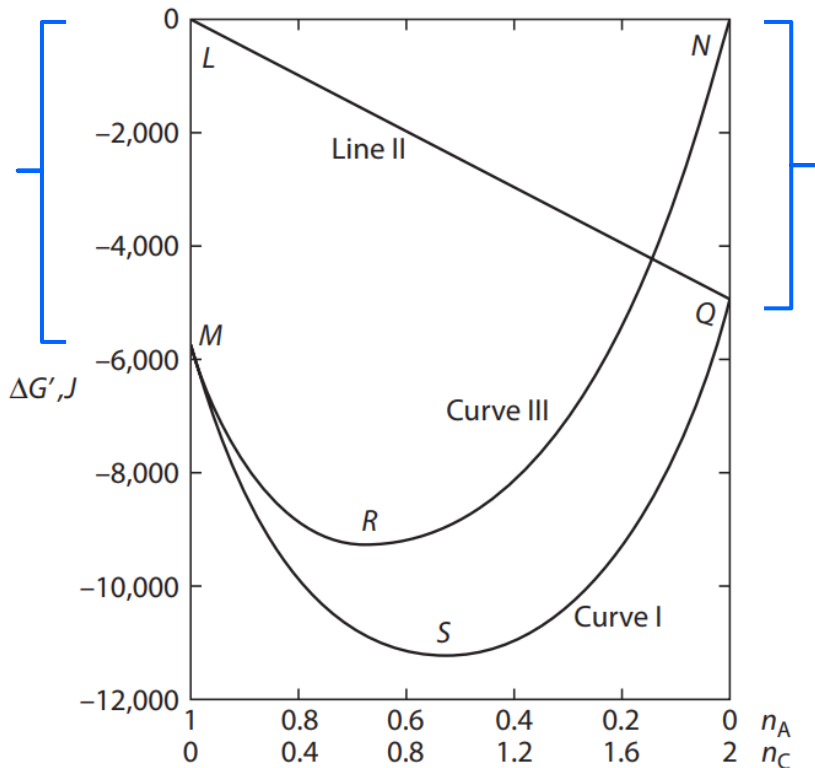
$$NQ = \Delta G^\circ = f(T)$$

- ✓ The effect of temperature on the composition at which the minimum occurs in **curve I** (and hence on the value of **K_p**) thus depends on the relative effects of temperature on the lengths of **LM** and **NQ** .

Figure 11.1

The decrease in **ΔG** due to **chemical reaction** (**line II**), due to gas **mixing** (**curve III**), Curve I is the sum of curves II and III.

11.3. The Effect of Temperature on the Equilibrium Constant



- ✓ For given reactants, the length of LM increases linearly with temperature, and the variation of the length NQ with temperature is determined by the sign and magnitude of the standard entropy change for the reaction according to

$$\left[\frac{\partial \Delta G^\circ}{\partial T} \right]_P = -\Delta S^\circ$$

Figure 11.1

The decrease in ΔG due to chemical reaction (line II), due to gas mixing (curve III), Curve I is the sum of curves II and III.

11.3. The Effect of Temperature on the Equilibrium Constant

- ✓ Increasing the temperature increases the length of LM , and
- ✓ if ΔS° is negative, increasing T must decrease ΔG° , hence decreasing the length of NQ .

$$\left[\frac{\partial \Delta G^\circ}{\partial T} \right]_P = -\Delta S^\circ$$

- ✓ Thus, the position of the minimum in the curve I shifts to the left, which indicates that K_p decreases with increasing temperature.
- ✓ The exact variation of K_p with temperature is obtained from consideration of the Gibbs–Helmholtz equation (Equation 5.38):

$$\left[\frac{\partial (\Delta G^\circ / T)}{\partial T} \right]_P = -\frac{\Delta H^\circ}{T^2}$$

11.3. The Effect of Temperature on the Equilibrium Constant

$$\Delta G^{\circ} = -RT \ln K_p$$

- ✓ From Equation 11.8, $\Delta G^{\circ} = -RT \ln K_p$, and since ΔG° is only a function of T ,

$$\frac{d \ln K_p}{dT} = -\frac{\Delta H^{\circ}}{RT^2}$$

- ✓ or

$$\frac{d \ln K_p}{d(1/T)} = -\frac{\Delta H^{\circ}}{R}$$

- ✓ Equation 11.10 is known as the *van't Hoff equation* (Jacobus Henricus van't Hoff, 1852–1911).

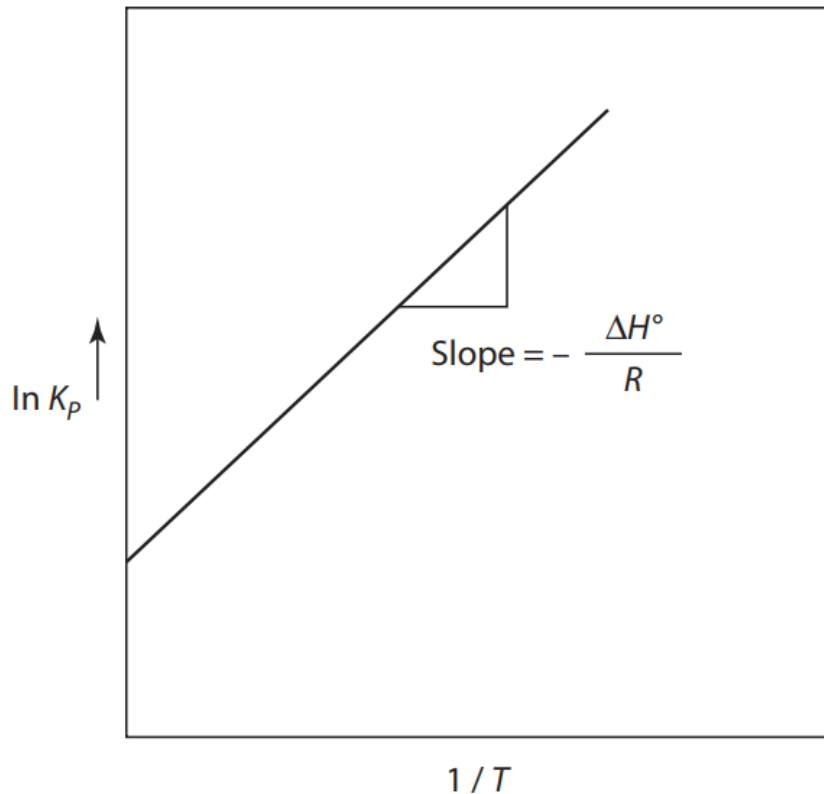
11.3. The Effect of Temperature on the Equilibrium Constant

- ✓ The equation shows that the effect of temperature on K_p is determined by the **sign** and **magnitude** of ΔH° for the reaction.

$$\frac{d \ln K_p}{d(1/T)} = -\frac{\Delta H^\circ}{R}$$

- If ΔH° is **positive**, the reaction is **endothermic** and K_p **increases** with increasing temperature.
- If ΔH° is **negative**, the reaction is **exothermic** and K_p **decreases** with increasing temperature.
- ✓ Integration of Equation 11.10 requires knowledge of the temperature dependence of ΔH° , which, as was seen in Chapter 6, depends on the value of Δc_p for the reaction.

11.3. The Effect of Temperature on the Equilibrium Constant



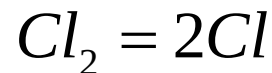
- ✓ The van't Hoff equation enables the determination of ΔH° from a plot of $\ln K_p$ versus the inverse of absolute temperature. The slope of the plot is $-(\Delta H^\circ / R)$ (Figure 11.2).

$$\frac{d \ln K_p}{d(1/T)} = -\frac{\Delta H^\circ}{R}$$

Figure 11.2 The variation of the $\ln(K_p)$ versus $1/T$ of an exothermic reaction (a van't Hoff plot).

11.3. The Effect of Temperature on the Equilibrium Constant

- ✓ The direction of the variation of K_p with temperature can be obtained from the application of Le Chatelier's principle (Henry Louis Le Chatelier, 1850–1936).
- ✓ If thermal energy is added to a system at reaction equilibrium, then the equilibrium is displaced in that direction which involves the absorption of heat.
- ✓ Consider the simple gaseous reaction

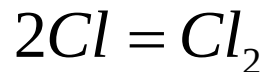


11.3. The Effect of Temperature on the Equilibrium Constant

- ✓ This reaction is **endothermic** and hence has a **positive** value of ΔH° .

Thus, the equilibrium constant, $K_P = p_{Cl}^2/p_{Cl_2}$, **increases** with increasing temperature; that is, the equilibrium shifts in that **direction** which involves the **absorption of thermal energy**.

- ✓ Conversely, if the reaction were written as



- ✓ then, since the reaction has a **negative** value of ΔH° , $K_P = p_{Cl_2}/p_{Cl}^2$ **decreases** with increasing temperature; that is, the equilibrium shifts in that **direction** which involves the **absorption of thermal energy**.

- ✓ In **both** cases, an increase in temperature **increases** p_{Cl} and **decreases** p_{Cl_2} .

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$$\left[\frac{p_C^2}{p_A p_B} \right]_{eq} = K_p$$

- ✓ The equilibrium constant, K_p , as defined by Equation 11.7, is independent of pressure.
- ✓ This is a consequence of the fact that ΔG° is the difference between the Gibbs free energies of the pure products (each at unit pressure) and the Gibbs free energies of the pure reactants (each at unit pressure), which is, by definition independent of pressure.
- ✓ However

11.4. The Effect of Pressure on the Equilibrium Constant

- ✓ However, if reaction equilibrium is expressed in terms of the number of moles of species present rather than in terms of the partial pressures of the species present, the constant is dependent on the total pressure if the chemical reaction involves a change in the total number of moles present.
- ✓ Consider again the reaction $\text{Cl}_2 = 2\text{Cl}$.
- ✓ Completion of this reaction causes a doubling of the number of moles present, and the effect of a change in pressure can again be obtained by the application of Le Chatelier's principle.

11.4. The Effect of Pressure on the Equilibrium Constant

- ✓ If the pressure exerted on a system at reaction equilibrium is **increased**, then the **equilibrium shifts** in that direction which **tends to decrease** the pressure exerted by the system; that is, it shifts in that direction which decreases the number of moles present.
- ✓ Specifically, the effect of pressure on the reaction equilibrium expressed in terms of the number of moles present (or in terms of mole fractions) can be seen as follows:

$$K_p = \frac{p_{Cl}^2}{p_{Cl_2}}, \text{ which is independent of pressure}$$
$$= \frac{X_{Cl}^2 P^2}{X_{Cl_2} P} = \frac{X_{Cl}^2 P}{X_{Cl_2}} = K_x P$$

11.4. The Effect of Pressure on the Equilibrium Constant

- ✓ where K_x is the equilibrium constant expressed in terms of the mole fractions.
- ✓ Thus, if the pressure is increased, then K_x decreases in order to maintain K_p constant, and the decrease in K_x is achieved by the reaction equilibrium shifting toward the Cl_2 side such that X_{Cl} decreases and X_{Cl_2} increases.
- ✓ In the case of the reaction $A + B = 2C$, the system at all times contains 2 moles of gas, and thus, the reaction equilibrium, expressed in terms of the mole fractions, is independent of pressure; that is,

$$K_p = \frac{p_C^2}{p_A p_B} = \frac{X_C^2 P^2}{(X_A P)(X_B P)} = \frac{X_C^2}{X_A X_B} = K_x$$

11.4. The Effect of Pressure on the Equilibrium Constant

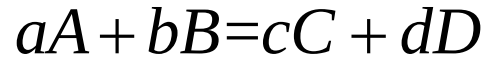
- ✓ This can also be seen from Equation 11.2, since, if $P \neq 1$, then the effect of the nonzero term $2RT \ln P$ is the raising or lowering of curve I in Figure 11.1 without affecting the position of the minimum with respect to the composition axis.

$$G' - 2G_C^\circ = n_A \left(-\Delta G^\circ \right) + 2RT \ln P \\ + 2RT \left[n_A \ln \left(\frac{n_A}{2} \right) + (1 - n_A) \ln (1 - n_A) \right]$$

- ✓ The magnitude of the effect of a change in pressure on the value of K_x depends on the magnitude of the change in the number of moles present in the system, occurring as a result of the chemical reaction.

11.4. The Effect of Pressure on the Equilibrium Constant

- ✓ For the general reaction



$$K_p = \frac{p_C^c p_D^d}{p_A^a p_B^b} = \frac{X_C^c X_D^d}{X_A^a X_B^b} \frac{P^c P^d}{P^a P^b} = K_x P^{(c+d-a-b)}$$

- ✓ which shows that K_x is **independent** of the total pressure **only if** $c + d = a + b$.

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11.8 Summary

11.5. Reaction Equilibrium as a Compromise between Enthalpy and Entropy

- ✓ Since the Gibbs free energy of a system is defined as

$$G' = H' - TS'$$

- ✓ low values of G are obtained with low values of H and large values of S .
- ✓ It was seen in the discussion of one-component systems in Chapter 7 that equilibrium occurs as the result of a compromise between enthalpy and entropy considerations. A similar discussion is now made concerning chemical reaction equilibria.
- ✓ Consider again the reaction $\text{Cl}_2 = 2\text{Cl}$. This reaction has a **positive** value of ΔH° (ΔH° is the thermal energy required to break Avogadro's number of $\text{Cl}-\text{Cl}$ bonds) and has a **positive** value of ΔS° (2 moles of chlorine atoms are produced from 1 mole of chlorine molecules).

11.5. Reaction Equilibrium as a Compromise between Enthalpy and Entropy

- ✓ Thus,
 - The system occurring as Cl atoms has a high value of H and a high value of S .
 - The system occurring as Cl_2 molecules has a low value of H and a low value of S .
- ✓ The minimum value of G thus occurs somewhere between the two extreme states.
- ✓ This compromise between the enthalpy and the entropy is analogous to the compromise between the contributions of the chemical reaction and the mixing of gases to the decrease in Gibbs free energy shown in Figure 11.1.

11.5. Reaction Equilibrium as a Compromise between Enthalpy and Entropy

- ✓ For the reaction $A + B = 2C$, Equation 11.3 can be written as

$$\begin{aligned} G' - 2G_C^\circ &= \underbrace{n_A(-\Delta H^0)}_{\text{chemical reaction}} + \underbrace{n_A(T\Delta S^0) + 2RT \left[n_A \ln\left(\frac{n_A}{2}\right) + (1-n_A)\ln(1-n_A) \right]}_{\text{Mixing}} \\ &= \left[n_A(-\Delta H^0) \right] + T \left\{ n_A\Delta S^0 + 2R \left[n_A \ln\left(\frac{n_A}{2}\right) + (1-n_A)\ln(1-n_A) \right] \right\} \end{aligned}$$

- ✓ The term in the first brackets is the enthalpy contribution to the change in the Gibbs free energy.
- ✓ The second term on the right-hand side of the equation is the entropy contribution, with $n_A\Delta S^\circ$ being the entropy change caused by the **chemical reaction** and $2R[n_A \ln(n_A/2) + (1-n_A) \ln(1-n_A)]$ being the entropy change caused by **gas mixing**.

11.5. Reaction Equilibrium as a Compromise between Enthalpy and Entropy

- ✓ In the example in Section 11.2, ΔG° equaled -5000 J at 500 K. Let it be that $\Delta H^\circ = -2500$ J and $\Delta S^\circ = 5$ J/K, in which case Figure 11.3 can be drawn from Figure 11.1.

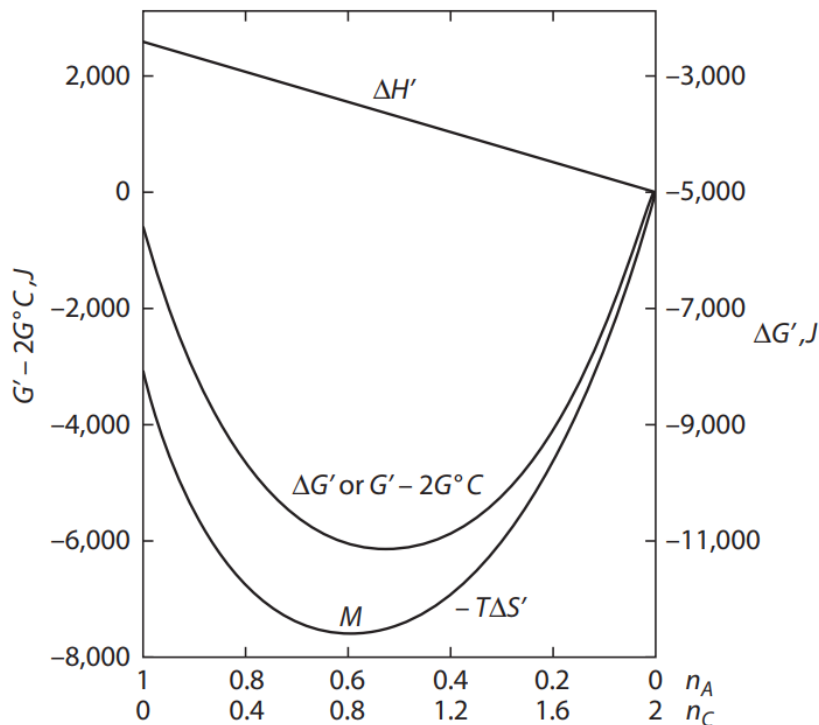


Figure 11.3

the variations at 500 K of $\Delta H'$, $-T\Delta S'$, and $\Delta G'$ with the extent of reaction $A_{(g)} + B_{(g)} = 2C_{(g)}$, for which $\Delta G^\circ = -2500 - 5T$.

11.5. Reaction Equilibrium as a Compromise between Enthalpy and Entropy

- ✓ In Figure 11.3, the $\Delta H'$ line is the first term on the right-hand side of Equation 11.11 and the $T\Delta S'$ curve is the second term.
- ✓ The sum of these two gives $G' - 2G_C^\circ$ the scale of which is given on the left-hand ordinate of Figure 11.3.
- ✓ The scale on the right-hand ordinate of the figure is $\Delta G'$, where, as before, the reference zero of Gibbs free energy is chosen as $G_A^\circ + G_B^\circ = 0$, such that $\Delta G' = G'$ (i.e., the scale is displaced by $2G_C^\circ = -5000 \text{ J}$).

11.5. Reaction Equilibrium as a Compromise between Enthalpy and Entropy

- ✓ On this scale, the $\Delta G'$ curve in Figure 11.3 is identical with curve I in Figure 11.1. It can be seen that the minimum in the $\Delta G'$ curve is determined as a compromise between the minimum value of H' at $n_A = 0$ and the maximum value of $T\Delta S'$ at $n_A = 0.597$ (the point M in Figure 11.2).
- ✓ If the temperature is increased, then the $T\Delta S'$ term becomes relatively more important, and hence the equilibrium value of n_A increases (the minimum in the $\Delta G'$ curve shifts to the left).
- ✓ Thus, K_p decreases with increasing temperature in accordance with Equation 11.9 for negative ΔH° .

11.5. Reaction Equilibrium as a Compromise between Enthalpy and Entropy

- ✓ The effect of temperature on the $\Delta G'$ curve is illustrated in Figure 11.4, in which curves are drawn for the reaction $A + B = 2C$ at 500, 1000, and 1500 K. It is assumed that Δc_p for the reaction is zero, in which case ΔH° and ΔS° are independent of temperature. Since $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$.

$$\Delta G_{500}^\circ = -2500 - (500 \times 5) = -5000 J \quad K_{p,500} = 3.329$$

$$\Delta G_{1000}^\circ = -2500 - (1000 \times 5) = -7500 J \quad K_{p,1000} = 2.465$$

$$\Delta G_{1500}^\circ = -2500 - (1500 \times 5) = -10000 J \quad K_{p,1500} = 2.229$$

- ✓ The equilibrium values of n_A at 500, 1000, and 1500 K are thus, respectively, 0.523, 0.560, and 0.572. Thus, increasing the temperature shifts the minimum to the left, thereby increasing the amount of A and B in equilibrium.

11.5. Reaction Equilibrium as a Compromise between Enthalpy and Entropy

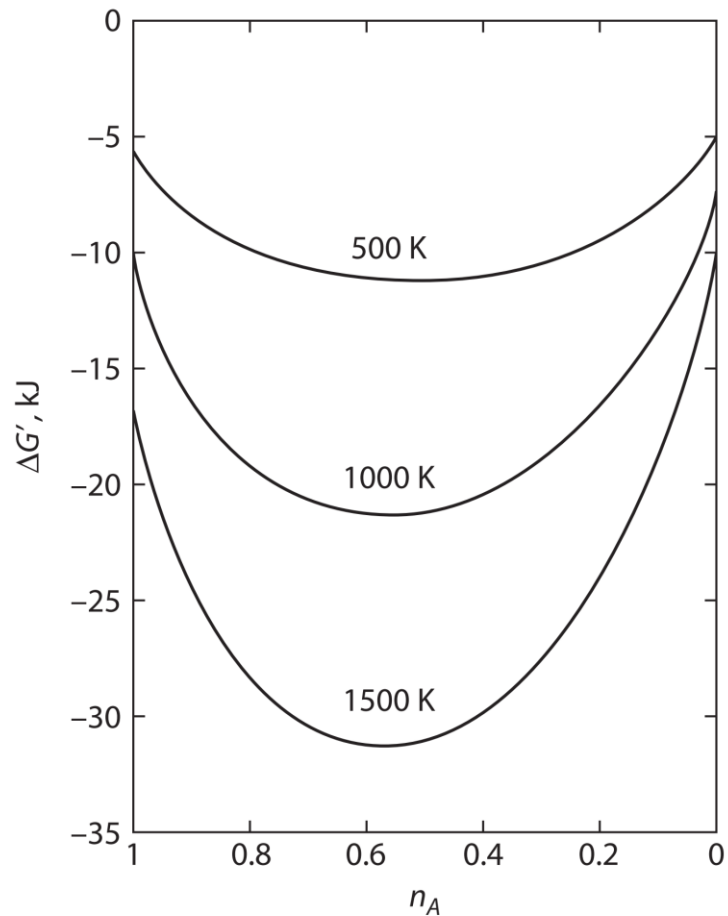


Figure 11.4 The effect of temperature on the equilibrium state of the reaction $A_{(g)} + B_{(g)} = 2C_{(g)}$, for which $\Delta G^\circ = -2500 - 5T$ J.

11.5. Reaction Equilibrium as a Compromise between Enthalpy and Entropy

- ✓ Although K_p is constant at constant temperature, note that an infinite set of partial pressures of reactants and products correspond to the fixed value of K_p .
- ✓ If the reaction involves three species, then an arbitrary choice of the partial pressures of two of the species uniquely fixes the equilibrium partial pressure of the third. This can also be shown by the Gibbs' equilibrium phase rule (see Qualitative Problem 2 in Section 11.10).

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11.8. Summary

1. The equilibrium state of the reaction $aA + bB = cC + dD$ is that in which $a\bar{G}_A + b\bar{G}_B = c\bar{G}_C + d\bar{G}_D$. The equilibrium state of the reaction is thus determined by the value of ΔG° for the reaction and is quantified by the equilibrium constant K_p , where $K_p = (p_C^c p_D^d / p_A^a p_B^b)_{eq}$ and the standard state for each reactant and product gas is the pure gas at 1 atm pressure.
2. ΔG° and K_p are related by $\Delta G^\circ = -RT \ln K_p$. This relation is one of the more powerful equations in chemical thermodynamics and will be used extensively in subsequent chapters. For increasingly negative values of ΔG° , K_p becomes increasingly greater than unity, and, conversely, for increasingly positive values of ΔG° , K_p becomes increasingly less than unity.

11.8. Summary

3. Since ΔG° is a function only of temperature, K_p is a function only of temperature, and the dependence of K_p on temperature is determined by the value of ΔH° for the reaction; that is,

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ = -RT \ln K_p$$

and hence,

$$\ln K_p = -\frac{\Delta H^\circ}{RT} + \frac{\Delta S^\circ}{R}$$

or

$$\frac{d \ln K_p}{dT} = \frac{\Delta H^\circ}{RT^2}$$

or

$$\frac{d \ln K_p}{d(1/T)} = -\frac{\Delta H^\circ}{R}$$

11.8. Summary

3. Thus, for an exothermic reaction, K_p decreases with increasing temperature, and for an endothermic reaction, K_p increases with increasing temperature.

11.8. Summary

4. Since $p_i = X_i P$ in an ideal gas mixture, the equilibrium constant can be written in terms of the mole fractions; that is,

$$K_p = \frac{P_C^c P_D^d}{P_A^a P_B^b} = \frac{X_C^c X_D^d}{X_A^a X_B^b} P^{(c+d-a-b)} = K_x P^{(c+d-a-b)}$$

Although K_p is, by definition, independent of pressure, K_x is independent of pressure only if $(c + d - a - b) = 0$ —that is, if the gas mixture contains a constant number of moles in all states along the reaction coordinate.

If the forward progression of the reaction decreases the number of moles present (i.e., if $[c + d - a - b] < 0$) then an increase in pressure increases the value of K_x ;

11.8. Summary

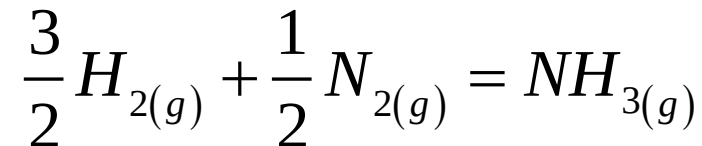
4. conversely, if $(c + d - a - b) > 0$, an increase in pressure decreases the value of K_x . The dependence of K_x on pressure and the dependence of K_p on temperature are examples of Le Chatelier' principle.

11.8. Summary

5. If ΔH° and ΔS° are known for the reaction, the following can be stated.
- If $\Delta H^\circ > 0$ and $\Delta S^\circ > 0$, the reaction will occur spontaneously at high temperatures.
 - If $\Delta H^\circ < 0$ and $\Delta S^\circ < 0$, the reaction will occur spontaneously at low temperatures.
 - If $\Delta H^\circ > 0$ and $\Delta S^\circ < 0$, the reaction will not occur spontaneously.
 - If $\Delta H^\circ < 0$ and $\Delta S^\circ > 0$, the reaction will occur spontaneously at all temperatures.

Homework

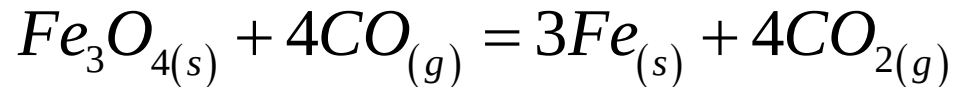
1. A gas mixture of 50% CO, 25% CO₂, and 25% H₂ (by volume) is fed to a furnace at 900°C. Determine the composition of the equilibrium CO–CO₂–H₂–H₂O gas if the total pressure of the gas in the furnace is 1 atm.
2. Ammonia gas is heated to 300°C. At what total pressure is the mole fraction of N₂ in the equilibrium gas mixture equal to 0.2? Calculate the standard enthalpy change and the standard entropy change for the reaction



at 300°C

Homework

3. A CO_2 – CO – H_2O – H_2 gas mixture at a total pressure of 1 atm exerts a partial pressure of oxygen of 10^{-7} atm at 1600 °C. In what ratio were the CO_2 and H_2 mixed to produce the gas with this oxygen pressure?
4. Consider the reduction of magnetite by CO to form pure iron:



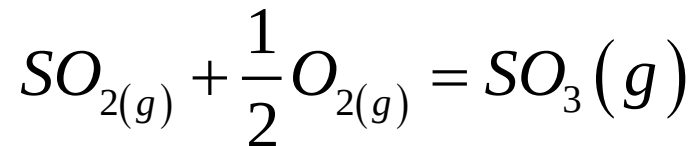
Calculate the partial pressure of CO and CO_2 in terms of the equilibrium constant K_P for the reaction

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- ✓11.8 Summary

11.6. Reaction Equilibrium in the System $\text{SO}_{2(g)}\text{-SO}_{3(g)}\text{-O}_{2(g)}$

- ✓ Consider the equilibrium reaction of the oxidation of sulfur dioxide to sulfur trioxide:



- ✓ The standard Gibbs free energy change for this reaction is

$$\Delta G^\circ = -94600 + 89.37T \text{ J}$$

- ✓ Thus, at 1000 K,

$$\Delta G_{1000}^\circ = -5230 \text{ J}$$

$$\ln K_p = \frac{5230}{8.3144 \times 1000} = 0.629$$

11.6. Reaction Equilibrium in the System $\text{SO}_{2(g)}-\text{SO}_{3(g)}-\text{O}_{2(g)}$

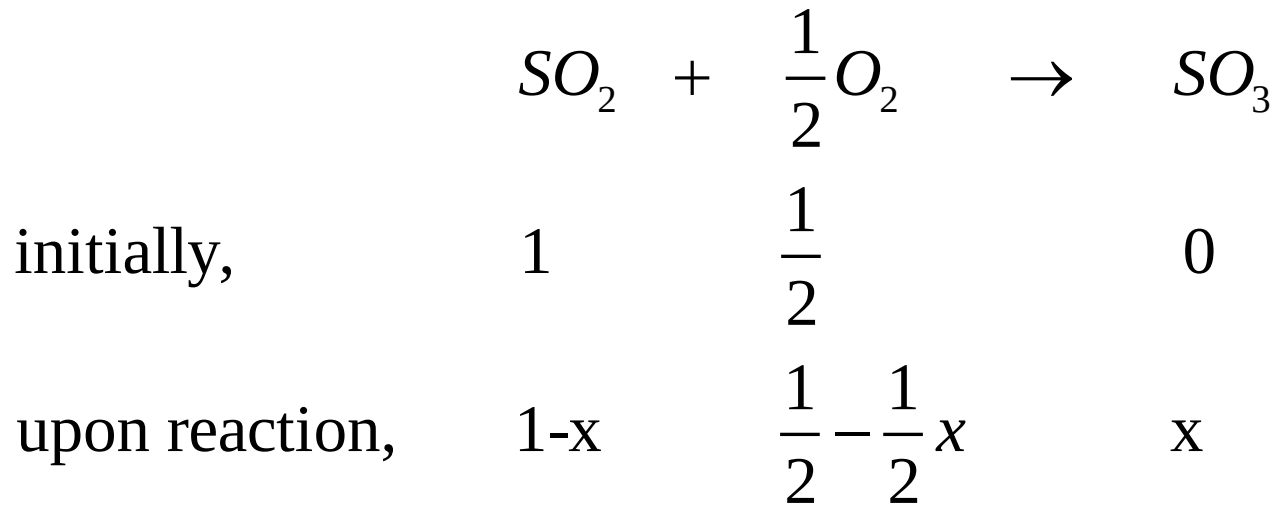
✓ and

$$K_p = 1.876 = \frac{p_{\text{SO}_3}}{p_{\text{SO}_2} p_{\text{O}_2}^{1/2}}$$

- ✓ Consider the reaction between 1 mole of SO_2 gas at 1 atm pressure and $\frac{1}{2}$ mole of O_2 gas at 1 atm pressure to form an equilibrium mixture of SO_2 , SO_3 , and O_2 at 1 atm pressure and 1000 K.
- ✓ From the stoichiometry of the chemical reaction given by Equation 11.12, x moles of SO_3 are formed from the reaction of x moles of SO_2 and $\frac{1}{2}x$ moles of O_2 .

11.6. Reaction Equilibrium in the System $\text{SO}_{2(g)}-\text{SO}_{3(g)}-\text{O}_{2(g)}$

- ✓ Thus, any reacting mixture contains x moles of SO_3 , $(1 - x)$ moles of SO_2 , and $\frac{1}{2}(1 - x)$ moles of O_2 ; that is,



- ✓ The total number of moles in the system, n_T , is

$$n_T = 1 - x + \frac{1}{2} - \frac{1}{2}x + x = \frac{1}{2}(3 - x)$$

11.6. Reaction Equilibrium in the System $\text{SO}_{2(g)}-\text{SO}_{3(g)}-\text{O}_{2(g)}$

✓ and, since

$$p_i = \frac{n_i}{n_T} P$$

✓ then

$$p_{\text{SO}_2} = \frac{2(1-x)P}{(3-x)}, \quad p_{\text{O}_2} = \frac{(1-x)P}{(3-x)}, \quad p_{\text{SO}_3} = \frac{2xP}{(3-x)}$$

✓ Thus,

$$K_p^2 = \frac{p_{\text{SO}_3}^2}{p_{\text{SO}_2}^2 p_{\text{O}_2}} = \frac{(3-x)x^2}{(1-x)^3 P}$$

11.6. Reaction Equilibrium in the System $\text{SO}_{2(g)}-\text{SO}_{3(g)}-\text{O}_{2(g)}$

✓ or

$$(1 - PK_p^2)x^3 + (3PK_p^2 - 3)x^2 - 3PK_p^2x + PK_p^2 = 0$$

✓ which, with $P = 1$ and $K_p = 1.876$, gives $x = 0.463$. Thus, at equilibrium, there are 0.537 moles of SO_2 , 0.269 moles of O_2 , and 0.463 moles of SO_3 , such that

$$p_{\text{SO}_2} = \frac{2 \times (1 - 0.463)}{(3 - 0.463)} = 0.423 \text{ atm}$$

$$p_{\text{O}_2} = \frac{(1 - 0.463)}{(3 - 0.463)} = 0.212 \text{ atm}$$

11.6. Reaction Equilibrium in the System $\text{SO}_{2(g)}-\text{SO}_{3(g)}-\text{O}_{2(g)}$

✓ and

$$p_{\text{SO}_3} = \frac{2 \times 0.463}{(3 - 0.463)} = 0.365 \text{ atm}$$

✓ As a check,

$$K_p = \frac{0.365}{0.423 \times 0.212^{1/2}} = 1.874$$

✓ The composition of the equilibrium gas is thus 42.3% SO_2 , 21.2% O_2 , and 36.5% SO_3 .

11.6. Reaction Equilibrium in the System $\text{SO}_{2(g)}-\text{SO}_{3(g)}-\text{O}_{2(g)}$

✓ 11.6.1. The Effect of Temperature

- ✓ Since ΔH° for the reaction given by Equation 11.12 is negative (–94,600 J), and Le Chatelier's principle predicts that a decrease in temperature at constant pressure shifts the equilibrium in that direction which involves an evolution of heat, decreasing the temperature at constant pressure causes the equilibrium to shift toward the SO_3 side.

- ✓ At 900K,

$$\Delta G_{900}^\circ = -14167 \text{ J}$$

$$\ln K_p = \frac{14167}{8.3144 \times 900} = 1.893$$

11.6. Reaction Equilibrium in the System $\text{SO}_{2(g)}-\text{SO}_{3(g)}-\text{O}_{2(g)}$

✓ 11.6.1. The Effect of Temperature

✓ and thus,

$$K_p = 6.64$$

✓ Substituting $K_p = 6.64$ and $P = 1$ atm into Equation 11.13 gives $x = 0.704$, and thus,

$$p_{\text{SO}_2} = 0.258 \text{ atm}, p_{\text{O}_2} = 0.129 \text{ atm}, p_{\text{SO}_3} = 0.613 \text{ atm}$$

✓ As a check,

$$K_p = \frac{0.613}{0.258 \times 0.129^{1/2}} = 6.64$$

11.6. Reaction Equilibrium in the System $\text{SO}_{2(g)}-\text{SO}_{3(g)}-\text{O}_{2(g)}$

✓ 11.6.1. The Effect of Temperature

- ✓ Thus, of the total number of moles present,
 - SO_2 constitutes 25.8%, which is a decrease from 42.3% at 1000 K.
 - O_2 constitutes 12.9%, which is a decrease from 21.2% at 1000 K.
 - SO_3 constitutes 61.3%, which is an increase from 36.5% at 1000 K.
- ✓ It is seen that a decrease in temperature has shifted the equilibrium toward the SO_3 side.

11.6. Reaction Equilibrium in the System $\text{SO}_{2(g)}-\text{SO}_{3(g)}-\text{O}_{2(g)}$

✓ 11.6.2. The Effect of Pressure

- ✓ Although K_p is independent of pressure, Le Chatelier's principle predicts that an increase in total pressure at constant temperature shifts the equilibrium in that direction which involves a decrease in the number of moles in the system—that is, toward the SO_3 side.
- ✓ Consider the equilibrium mixture at $P = 10$ atm and $T = 1000$ K. Substituting $P = 10$ and $K_p = 1.876$ into Equation 11.13 gives $x = 0.686$, in which case $n_{\text{SO}_2} = 0.314$, $n_{\text{O}_2} = 0.157$, $n_{\text{SO}_3} = 0.686$. Thus,

11.6. Reaction Equilibrium in the System $\text{SO}_{2(\text{g})}$ - $\text{SO}_{3(\text{g})}$ - $\text{O}_{2(\text{g})}$

✓ 11.6.2. The Effect of Pressure

$$p_{\text{SO}_2} = \frac{2 \times (1 - 0.686) \times 10}{(3 - 0.686)} = 2.714 \text{ atm}$$

$$p_{\text{O}_2} = \frac{(1 - 0.686) \times 10}{(3 - 0.686)} = 1.357 \text{ atm}$$

✓ and

$$p_{\text{SO}_3} = \frac{2 \times 0.686 \times 10}{(3 - 0.686)} = 5.929 \text{ atm}$$

✓ As a check,

$$K_p = \frac{5.929}{2.714 \times 1.357^{1/2}} = 1.875$$

11.6. Reaction Equilibrium in the System $\text{SO}_{2(g)}-\text{SO}_{3(g)}-\text{O}_{2(g)}$

✓ 11.6.2. The Effect of Pressure

- ✓ Of the total number of moles present,
 - SO_2 constitutes 27.1%, which is a decrease from 42.3% at $P = 1$ atm.
 - O_2 constitutes 13.75%, which is a decrease from 21.2% at $P = 1$ atm.
 - SO_3 constitutes 59.29%, which is an increase from 36.5% at $P = 1$ atm.
- ✓ Thus, it is seen that an increase in pressure has shifted the equilibrium toward the SO_3 side.

11.6. Reaction Equilibrium in the System $\text{SO}_{2(g)}-\text{SO}_{3(g)}-\text{O}_{2(g)}$

✓ 11.6.3. The Effect of Changes in Temperature and Pressure

- ✓ In order to simultaneously vary the temperature and the pressure in such a manner that the numbers of moles of the three gaseous species present remain constant, Equation 11.13 indicates that the variation must be such that the term PK_p^2 remains constant, where

$$K_p = \exp\left(-\frac{\Delta G^0}{RT}\right) = \exp\left(\frac{94600}{8.3144 \cdot T}\right) \cdot \exp\left(-\frac{89.37}{8.3144}\right)$$

- ✓ It is apparent that, by mixing SO_2 gas and SO_3 gas, an equilibrium mixture with a known partial pressure of O_2 can be produced.

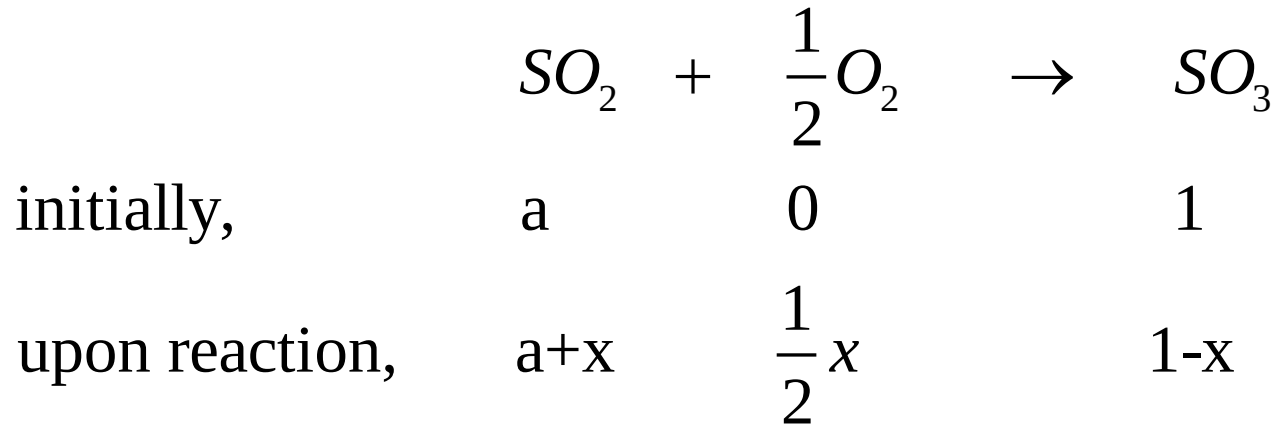
11.6. Reaction Equilibrium in the System $\text{SO}_{2(g)}-\text{SO}_{3(g)}-\text{O}_{2(g)}$

✓ 11.6.3. The Effect of Changes in Temperature and Pressure

- ✓ For example, consider that it was required to have an $\text{SO}_3-\text{SO}_2-\text{O}_2$ mixture at 1 atm total pressure in which $p_{\text{O}_2} = 0.1 \text{ atm}$.
- ✓ To obtain this gas mixture, SO_3 and SO_2 , both at 1 atm pressure, would be mixed in the molar ratio $\text{SO}_2/\text{SO}_3 = a$ and allowed to equilibrate.
- ✓ If a moles of SO_2 and 1 mole of SO_3 are mixed, then from the stoichiometry of Equation 11.12, x moles of SO_3 would decompose to form x moles of SO_2 and $\frac{1}{2}x$ moles of O_2 , such that, at equilibrium, the number of moles present would be

11.6. Reaction Equilibrium in the System $\text{SO}_{2(g)}-\text{SO}_{3(g)}-\text{O}_{2(g)}$

✓ 11.6.3. The Effect of Changes in Temperature and Pressure



✓ with

$$n_T = a + x + \frac{1}{2}x + 1 - x = \frac{1}{2}(2a + x + 2)$$

✓ In this gas mixture,

$$p_{\text{O}_2} = \frac{n_{\text{O}_2}}{n_T} = \frac{x}{2a + 2 + x} P$$

11.6. Reaction Equilibrium in the System $\text{SO}_{2(g)}-\text{SO}_{3(g)}-\text{O}_{2(g)}$

✓ 11.6.3. The Effect of Changes in Temperature and Pressure

$$p_{\text{O}_2} = \frac{n_{\text{O}_2}}{n_T} = \frac{x}{2a + 2 + x} P$$

✓ which, for $P = 1$ and $p_{\text{SO}_2} = 0.1$, gives $a = 4.5x - 1$. Also,

$$K_p^2 = \frac{p_{\text{SO}_3}^2}{p_{\text{SO}_2}^2 p_{\text{O}_2}} = \frac{(1-x)^2 (2a+2+x)}{(a+x)^2 xP}$$

✓ At 1000 K, $K_p = 1.876$. Thus, substituting for a in terms of x and taking $K_p^2 = 3.519$ gives

$$96.45x^3 - 18.709x^2 - 6.481x = 0$$

11.6. Reaction Equilibrium in the System $\text{SO}_{2(g)}-\text{SO}_{3(g)}-\text{O}_{2(g)}$

✓ 11.6.3. The Effect of Changes in Temperature and Pressure

✓ or, since $x \neq 0$,

$$96.45x^2 - 18.709x - 6.481 = 0$$

✓ which has the solution $x = 0.374$. Thus,

$$a = (4.5 \times 0.374) - 1 = 0.683$$

$$p_{\text{O}_2} = \frac{xP}{2a+2+x} = \frac{0.374}{(2 \times 0.683) + 2 + 0.374} = 0.1 \text{ atm}$$

$$p_{\text{SO}_3} = \frac{2(1-x)}{2a+2+x} = \frac{2(1-0.374)}{(2 \times 0.683) + 2 + 0.374} = 0.335 \text{ atm}$$

11.6. Reaction Equilibrium in the System $\text{SO}_{2(g)}-\text{SO}_{3(g)}-\text{O}_{2(g)}$

✓ 11.6.3. The Effect of Changes in Temperature and Pressure

✓ and

$$p_{\text{SO}_2} = \frac{2(a+x)}{2a+2+x} = \frac{2(0.683+0.374)}{(2 \times 0.683) + 2 + 0.374} = 0.565 \text{ atm}$$

✓ In the equilibrium mixture, $p_{\text{SO}_2}/p_{\text{SO}_3} = 1.7$, compared with $p_{\text{SO}_2}/p_{\text{SO}_3} = a = 0.683$ in the initial mixture. As a check,

$$K_p = \frac{0.355}{0.565 \times 0.1^{1/2}} = 1.875$$

11.6. Reaction Equilibrium in the System $\text{SO}_{2(g)}-\text{SO}_{3(g)}-\text{O}_{2(g)}$

✓ 11.6.3. The Effect of Changes in Temperature and Pressure

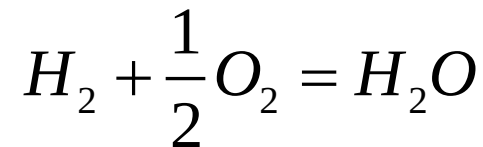
- ✓ If it had been required to have $p_{\text{O}_2} = 0.212$ atm (the equilibrium value of p_{O_2} in an equilibrated gas at 1 atm pressure of initial composition 1 mole of $\text{SO}_2 + \frac{1}{2}$ mole of O_2), then the solution of the stoichiometric condition would have given $a = 0$, which indicates that pure SO_3 at 1000 K and 1 atm pressure decomposes to the same equilibrium gas mixture as that formed from an initial mixture of 1 mole of $\text{SO}_2 + \frac{1}{2}$ mole of O_2 at 1 atm. Thus, 0.212 atm is the maximum value of p_{O_2} which can be produced in an SO_2-SO_3 mixture at 1000 K and 1 atm total pressure.

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11.7. Equilibrium in H₂O-H₂ and CO₂ Mixtures

- ✓ H₂O-H₂ and CO₂-CO gas mixtures are used when it is required that the partial pressure of oxygen in a gas phase be fixed at a very low value.
- ✓ For example, if it were required to have a gaseous atmosphere containing a partial pressure of oxygen of 10⁻¹⁰ atm, then such an oxygen potential can be obtained with relative ease by establishing the equilibrium



- ✓ for which

$$\Delta G^\circ = -247500 + 55.85T \text{ J}$$

11.7. Equilibrium in H₂O-H₂ and CO₂ Mixtures

✓ From Equation 11.8,

$$\ln K_p = \frac{247500}{8.3144 \cdot T} - \frac{55.85}{8.3144}$$

✓ If it is required to have an atmosphere containing $p_{O_2} = 10^{-10}$ atm at $T = 2000$ K, then, at this temperature,

$$\ln K_p = \frac{247500}{8.3144 \cdot 2000} - \frac{55.85}{8.3144} = 8.167$$

✓ Therefore,

$$K_p = 3.521 \times 10^3 = \frac{p_{H_2O}}{p_{H_2} p_{O_2}^{1/2}}$$

11.7. Equilibrium in H₂O-H₂ and CO₂ Mixtures

- ✓ and with $p_{O_2} = 10^{-10}$ atm,

$$\frac{p_{H_2O}}{p_{H_2}} = 3.521 \times 10^3 \times 10^{-5} = 3.521 \times 10^{-2}$$

- ✓ Thus, in the H₂-H₂O gas mixture, if $p_{H_2} = 1$ atm, then p_{H_2O} must be 0.0352 atm.

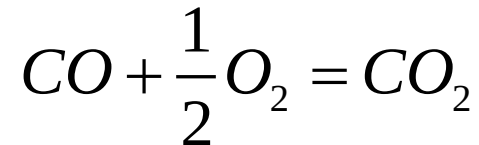
- ✓ The saturated vapor pressure of liquid water at 27.0 °C is found from

$$\log p(\text{atm}) = -\frac{2900}{T} - 4.65 \cdot \log T + 19.732$$

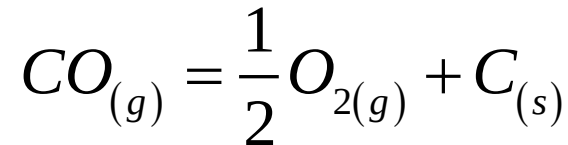
- ✓ to be 0.0352. Thus, the required gas mixture can be produced by bubbling hydrogen gas at 1 atm pressure through pure liquid water at 27.0 °C to saturate it with water vapor.

11.7. Equilibrium in H₂O-H₂ and CO₂ Mixtures

- ✓ The establishment of reaction equilibrium at 2000 K gives $p_{O_2} = 10^{-10}$ atm in the gas.
- ✓ Similarly, the partial pressure of oxygen in a gaseous atmosphere can be determined by establishing the reaction



- ✓ The standard Gibbs free energy change for the reaction

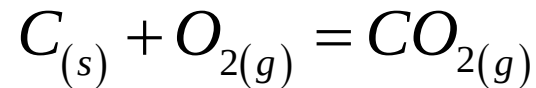


- ✓ is

$$\Delta G^\circ = 111700 + 87.65T \text{ J}$$

11.7. Equilibrium in H₂O-H₂ and CO₂ Mixtures

✓ and that for the reaction



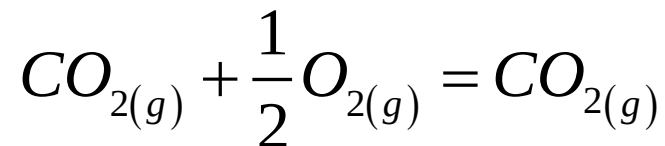
✓ is

$$\Delta G^{\circ} = -394100 - 0.84T \text{ J}$$

✓ Summation of the standard Gibbs free energy changes gives

$$\Delta G^{\circ} = -282400 + 86.81T \text{ J}$$

✓ for the reaction



11.7. Equilibrium in H₂O-H₂ and CO₂ Mixtures

✓ Thus,

$$\ln K_p = \frac{282400}{8.3144 \cdot T} - \frac{86.81}{8.3144}$$

✓ If it were required to have $p_{O_2} = 10^{-20}$ atm at 1000K, then

$$\ln K_p = \frac{282400}{8.3144 \cdot 1000} - \frac{86.81}{8.3144} = 23.52$$

✓ Therefore,

$$K_p = 1.646 \times 10^{10} = \frac{p_{CO_2}}{p_{CO} p_{O_2}^{1/2}} = \frac{p_{CO_2}}{p_{CO} \cdot 10^{-10}}$$

✓ and so

$$\frac{p_{CO_2}}{p_{CO}} = 1.646$$

11.7. Equilibrium in $\text{H}_2\text{O}-\text{H}_2$ and CO_2 Mixtures

- ✓ If the total pressure $P = 1$ atm, then, since $p_{\text{CO}_2} + p_{\text{CO}} = 1$, the partial pressure, $p_{\text{CO}_2} = 1.646(1 - p_{\text{CO}_2}) = 0.622$ atm and $p_{\text{CO}} = 0.378$. The required mixture is produced by mixing CO_2 and CO in the volume ratio 1.561/1—that is, 62.2 volume percent CO_2 and 37.8 volume percent CO .
- ✓ In both of the preceding equilibria, the oxygen pressure in the equilibrated gas is so small that the $p_{\text{H}_2}/p_{\text{H}_2\text{O}}$ and $p_{\text{CO}_2}/p_{\text{CO}}$ ratios in the equilibrated gases are negligibly different from the corresponding ratios in the initial mixtures.
- ✓ The equality of the ratio of the volume percentages with the ratio of the partial pressures in a gas mixture can be demonstrated as follows.

11.7. Equilibrium in H₂O-H₂ and CO₂ Mixtures

- ✓ Consider a cm³ of gas A at 1 atm pressure and b cm³ of gas B at 1 atm pressure being mixed at constant pressure (and hence at constant total volume, $a + b$).
- ✓ The number of moles of $A = n_A = (1+a)/RT = a/RT$, and the number of moles of $B = n_B = (1 \times b)/RT = b/RT$. Therefore, in the mixture,

$$p_A = \frac{n_A RT}{V'} = \frac{n_A RT}{(a + b)}$$

- ✓ and

$$p_B = \frac{n_B RT}{(a + b)}$$

- ✓ Thus,
$$\frac{p_A}{p_B} = \frac{n_A}{n_B} = \frac{a}{b} = \frac{\text{volume percentage of A}}{\text{volume percentage of B}}$$

Thermodynamics of Materials & Phase transformations

The End

